Software Requirements Specification

for

Vibecheck

Version 1.0 approved

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Revision History

Name	Date	Reason For Changes	Version

1. Introduction

1.1 Purpose

The Software Requirements Specification (SRS) defines the functional and non-functional requirements for Vibecheck, a web-based gaming companion application. The website aims to help users better understand their personality through an assessment and provide a personalised experience to discover new games and match with other users to play games.

1.2 Document Conventions

This document follows a consistent hierarchical numbering scheme for sections and requirements. Functional and non-functional requirements are uniquely numbered to support identification and traceability throughout the development process.

The keywords **"shall"** and **"must"** indicate mandatory system requirements, while **"may"** indicates optional behaviour. Technical and domain-specific terms used throughout the document are defined consistently in the Data Dictionary.

Requirements are written as atomic, precise and verifiable statements wherever possible.

1.3 Intended Audience and Reading Suggestions

This SRS is intended for the project team, the course coordinators and TA of this course. As well as other stakeholders related to this project.

Readers should familiarise themselves with the Big 5 Personality Traits, which is the base of how our algorithm works for Games recommendation and Gamer matching.

1.4 Product Scope

VibeCheck is a web-based platform designed to help users to understand their personality better through gaming. At the same time, they can discover new games and match with players similar to them..

The system allows users to create an account and complete a personality assessment that produces a gaming preference profile. This profile is used to provide personalized game recommendations and assist users in finding compatible gamers. The system also provides features for monitoring gaming behaviour to encourage greater awareness of gaming habits.

VibeCheck aims to reduce the difficulty users face when deciding what games to play or finding compatible people to play with through personality-based profiling.

1.5 References

Understanding Big Five Personality – https://www.verywellmind.com/the-big-five-personality-dimensions-2795422

Big Five Personality Test – https://bigfive-test.com/

Video Game API – https://rawg.io/

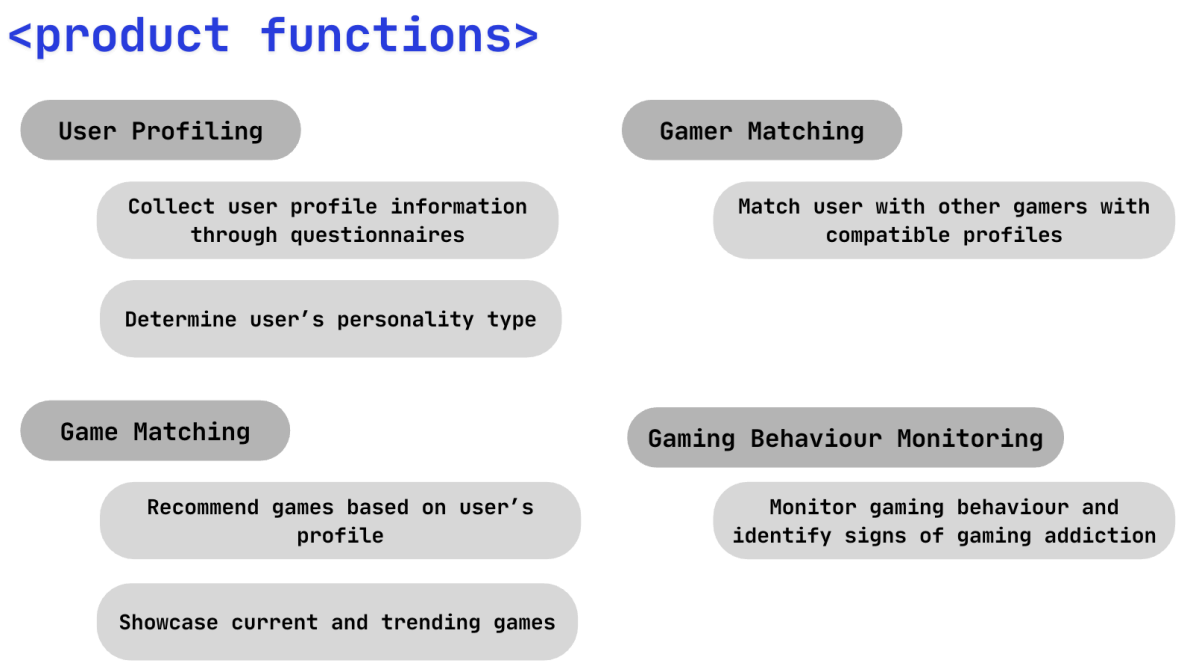
Game Recommendation Website (Match you with games based on 3 game titles given) – https://apps.quanticfoundry.com/recommendations/gamerprofile/videogame/

2. Overall Description

2.1 Product Perspective

<Describe the context and origin of the product being specified in this SRS. For example, state whether this product is a follow-on member of a product family, a replacement for certain existing systems, or a new, self-contained product. If the SRS defines a component of a larger system, relate the requirements of the larger system to the functionality of this software and identify interfaces between the two. A simple diagram that shows the major components of the overall system, subsystem interconnections, and external interfaces can be helpful.>

2.2 Product Functions



<Summarize the major functions the product must perform or must let the user perform. Details will be provided in Section 3, so only a high level summary (such as a bullet list) is needed here. Organize the functions to make them understandable to any reader of the SRS. A picture of the major groups of related requirements and how they relate, such as a top level data flow diagram or object class diagram, is often effective.>

2.3 User Classes and Characteristics

<Identify the various user classes that you anticipate will use this product. User classes may be differentiated based on frequency of use, subset of product functions used, technical expertise, security or privilege levels, educational level, or experience. Describe the pertinent characteristics of each user class. Certain requirements may pertain only to certain user classes. Distinguish the most important user classes for this product from those who are less important to satisfy.>

2.4 Operating Environment

<Describe the environment in which the software will operate, including the hardware platform, operating system and versions, and any other software components or applications with which it must peacefully coexist.>

2.5 Design and Implementation Constraints

<Describe any items or issues that will limit the options available to the developers. These might include: corporate or regulatory policies; hardware limitations (timing requirements, memory requirements); interfaces to other applications; specific technologies, tools, and databases to be used; parallel operations; language requirements; communications protocols; security considerations; design conventions or programming standards (for example, if the customer's organization will be responsible for maintaining the delivered software).>

2.6 User Documentation

<List the user documentation components (such as user manuals, on-line help, and tutorials) that will be delivered along with the software. Identify any known user documentation delivery formats or standards.>

2.7 Assumptions and Dependencies

<List any assumed factors (as opposed to known facts) that could affect the requirements stated in the SRS. These could include third-party or commercial components that you plan to use, issues around the development or operating environment, or constraints. The project could be affected if these assumptions are incorrect, are not shared, or change. Also identify any dependencies the project has on external factors, such as software components that you intend to reuse from another project, unless they are already documented elsewhere (for example, in the vision and scope document or the project plan).>

3. External Interface Requirements

3.1 User Interfaces

<Describe the logical characteristics of each interface between the software product and the users. This may include sample screen images, any GUI standards or product family style guides that are to be followed, screen layout constraints, standard buttons and functions (e.g., help) that will appear on every screen, keyboard shortcuts, error message display standards, and so on. Define the software components for which a user interface is needed. Details of the user interface design should be documented in a separate user interface specification.>

3.2 Hardware Interfaces

<Describe the logical and physical characteristics of each interface between the software product and the hardware components of the system. This may include the supported device types, the

nature of the data and control interactions between the software and the hardware, and communication protocols to be used.>

3.3 Software Interfaces

<Describe the connections between this product and other specific software components (name and version), including databases, operating systems, tools, libraries, and integrated commercial components. Identify the data items or messages coming into the system and going out and describe the purpose of each. Describe the services needed and the nature of communications. Refer to documents that describe detailed application programming interface protocols. Identify data that will be shared across software components. If the data sharing mechanism must be implemented in a specific way (for example, use of a global data area in a multitasking operating system), specify this as an implementation constraint.>

3.4 Communications Interfaces

<Describe the requirements associated with any communications functions required by this product, including e-mail, web browser, network server communications protocols, electronic forms, and so on. Define any pertinent message formatting. Identify any communication standards that will be used, such as FTP or HTTP. Specify any communication security or encryption issues, data transfer rates, and synchronization mechanisms.>

4. System Features

4.1 User Profile

4.1.1 Description and Priority

Description:

The User Profiling feature allows users to complete a questionnaire that collects information about their gaming preferences and personality. An algorithm will then process this information to generate the user's personality type and traits. This profile is also used for gamer matching and game recommendations.

Priority: High

4.1.2 Stimulus/Response Sequences

Stimulus	Response
User selects Create Gamer Profile	System displays questionnaire
User answers a questionnaire question	System records the selected response
User attempts to continue without answering a required question	System informs the user that the question must be answered
User submits the completed questionnaire	System validates and stores the questionnaire responses
Questionnaire is successfully submitted	System generates the user's personality type
Personality profile has been calculated	System displays the user's personality type/traits
User chooses to retake the questionnaire	System allows the user to submit a new set of responses and updates the profile

4.1.3 Functional Requirements

FR-1.1	The system shall allow the user to start the questionnaire
FR-1.2	The system shall display the questions contained in the questionnaire.
FR-1.3	The system shall allow the user to provide a response to each question.

FR-1.4	The system shall validate that all questionnaire questions have been answered before allowing the user to submit.
FR-1.5	The system shall store the user's submitted questionnaire responses.
FR-1.6	The system shall generate the user's personality type and traits based on the submitted responses.
FR-1.7	The system shall store the user's profile.
FR-1.8	The system shall display the personality profile to the user.
FR-1.9	The system shall allow the user to retake the questionnaire.
FR-1.10	The system shall update the user's profile after a new questionnaire is submitted.

4.2 Gamer Matching

4.2.1 Description and Priority

Description:

The Gamer Matching feature matches the user with other gamers with compatible gamer's profile. It uses information such as the user's personality profile and gaming preferences to determine the compatibility.

Priority: High

4.2.2 Stimulus/Response Sequences

Stimulus	Response
User selects the gamer matching feature	System retrieves the user's gamer profile
User requests gamer matches	System compares the user's profile with other gamer profiles
Compatible gamers are found	System displays a list of compatible gamers
User selects a recommended gamer	System displays the selected gamer's available profile information
No compatible gamers are found	System informs the user that no suitable matches are currently available

4.2.3 Functional Requirements

FR-2.1	The system shall allow the user to request gamer recommendations.
FR-2.2	The system shall retrieve the user's gamer profile when generating gamer recommendations.
FR-2.3	The system shall compare the user's gamer profile with other gamer profiles for compatibility.
FR-2.4	The system shall determine the compatibility between the user and other gamers.
FR-2.5	The system shall generate a list of compatible gamers.
FR-2.6	The system shall display the recommended gamers to the user.
FR-2.7	The system shall allow the user to select a recommended gamer.
FR-2.8	The system shall display the available profile information of the selected gamer.
FR-2.9	The system shall inform the user when no compatible gamers are available.

4.3 Game Recommendation

4.3.1 Description and Priority

Description:

The Game Recommendation feature recommends games to users based on their personality profile and gaming preferences. This feature also allows users to discover current trending games.

Priority: High

4.3.2 Stimulus/Response Sequences

Stimulus	Response
User selects the game recommendation feature	System retrieves the user's gamer profile
User requests personalized recommendations	System generates games based on the user's profile
Recommendations are generated	System displays the recommended games
User selects a recommended game	System displays information about the selected game
User selects trending games	System displays currently trending games
No suitable recommendations are available	System informs the user that no suitable recommendations were found, display trending games instead

4.3.3 Functional Requirements

FR-3.1	The system shall allow the user to request game recommendations.
FR-3.2	The system shall retrieve the user's gamer profile when generating game recommendations
FR-3.3	The system shall generate recommendations based on the user's personality profile and gaming preferences.
FR-3.4	The system shall display the recommended games to the user.
FR-3.5	The system shall allow the user to select a recommended game.
FR-3.6	The system shall display information about the selected game.
FR-3.7	The system shall allow the user to view currently trending games.
FR-3.8	The system shall display a list of currently trending games.
FR-3.9	The system shall inform the user if no suitable game recommendations are available and display trending games instead.

4.4 Game Behaviour Monitoring

4.4.1 Description and Priority

Description:

The Gaming Behaviour Monitoring feature records and analyses the user's gaming behaviour to identify behaviours that may signal excessive gaming. The system then sends a notification to the user regarding their behaviour and encourages them to take a break.

Priority: Medium

4.4.2 Stimulus/Response Sequences

Stimulus	Response
Gaming activity data is recorded	System stores the relevant gaming activity information
New gaming activity is available	System updates the user's gaming behaviour records
System analyses the user's gaming behaviour	System evaluates the behaviour for warning signs
Potentially excessive gaming behaviour is identified	System displays a notification to the user
User requests their gaming behaviour information	System displays a summary of the user's gaming activity

4.4.3 Functional Requirements

FR-4.1	The system shall record relevant gaming activity data for the user.
FR-4.2	The system shall maintain a history of the user's recorded gaming activity.
FR-4.3	The system shall analyse the user's recorded gaming behaviour.
FR-4.4	The system shall evaluate the user's gaming behaviour against warning signs of potentially excessive gaming.
FR-4.5	The system shall identify when the user's gaming behaviour satisfies the predefined warning signs.
FR-4.6	The system shall notify the user when potentially excessive gaming behaviour is identified.

FR-4.7	The system shall allow the user to view a summary of their gaming behaviour.

5. Other Nonfunctional Requirements

5.1 Performance Requirements

NFR-1.1	Data Retrieval Efficiency - All pages should be loaded within 3 seconds under normal network conditions. - All game recommendations and player matching should be loaded within 5 seconds
NFR-1.2	Constant Visible Feedback - All user interface interactions shall receive visible feedback within 1 second.
NFR-1.3	Concurrency Support - The system shall support at least 30 concurrent users while satisfying all performance requirements stated above.

5.2 Safety Requirements

<Specify those requirements that are concerned with possible loss, damage, or harm that could result from the use of the product. Define any safeguards or actions that must be taken, as well as actions that must be prevented. Refer to any external policies or regulations that state safety issues that affect the product's design or use. Define any safety certifications that must be satisfied.>

NFR-2.1	The system shall not recommend a game to users whose registered age is below the game's minimum age rating.
NFR-2.2	The system shall not recommend games with unverified or incomplete information.
NFR-2.3	System shall not recommend game contents involving harassment, hate speech, or sexual material.
NFR-2.4	Notifications of excessive gaming behaviour shall be in neutral and non-judgmental language. Strictly no blame or threaten user.

5.3 Security Requirements

<Specify any requirements regarding security or privacy issues surrounding use of the product or protection of the data used or created by the product. Define any user identity authentication requirements. Refer to any external policies or regulations containing security issues that affect the product. Define any security or privacy certifications that must be satisfied.>

NFR-3.1	Passwords shall not be stored in plain text and shall be encrypted using hashing algorithms.
NFR-3.2	Users shall be able to block another user. A blocked user shall not be able to match with the blocking user.
NFR-3.3	Private information shall not be used in gamer matching without user consent.
NFR-3.4	Users shall be able to request deletion of their account and personal data.

5.4 Software Quality Attributes

5.4.1 Usability

NFR-4.1	New user shall be able to complete the questionnaire and view their results directly on the application.
NFR-4.2	The interface shall apply consistent navigation, terminology, theme, and interaction patterns throughout the application
NFR-4.3	The interface shall be responsive on desktop, tablet, and mobile screens.
NFR-4.4	Content shall be displayed in the user's languages according to the user's locale.

5.4.2 Accessibility

NFR-4.4	System's interface shall have sufficient contrast ratios for text, icons, and background.
NFR-4.5	All interactive elements shall have clear labels.

5.4.3 Maintainability

NFR-4.6	Changes in configuration, such as reminder intervals, shall not require changes to source code.
NFR-4.7	Core functionalities shall be implemented as separable modules.

5.5 Business Rules

<List any operating principles about the product, such as which individuals or roles can perform which functions under specific circumstances. These are not functional requirements in themselves, but they may imply certain functional requirements to enforce the rules.>

NFR-5.1	User must complete the minimum required personality questionnaire before receiving personalized game recommendations or gamer matches.
NFR-5.2	Personal profile shall be recalculated when and only when the user retakes the questionnaire.
NFR-5.3	Users shall be paired only with gamers who have enable gamer matching feature and have not blocked each other.
NFR-5.4	All user information shall be private and not share to other users by default.
NFR-5.5	Users' gaming behaviour information shall be private and not be shared.
NFR-5.6	All data collected from users shall not be shared for other purposes.

6. Other Requirements

<Define any other requirements not covered elsewhere in the SRS. This might include database requirements, internationalization requirements, legal requirements, reuse objectives for the project, and so on. Add any new sections that are pertinent to the project.>

Appendix A: Glossary

<Define all the terms necessary to properly interpret the SRS, including acronyms and abbreviations. You may wish to build a separate glossary that spans multiple projects or the entire organization, and just include terms specific to a single project in each SRS.>

Appendix B: Analysis Models

<Optionally, include any pertinent analysis models, such as data flow diagrams, class diagrams, state-transition diagrams, or entity-relationship diagrams.>

Appendix C: To Be Determined List

<Collect a numbered list of the TBD (to be determined) references that remain in the SRS so they can be tracked to closure.>

Source: http://www.frontiernet.net/~kwiegers/process_assets/srs_template.doc